

Experiment No. 4

Title:

Use function types, scope, and closures; apply try-catch; build a palindrome checker.

Software / Tools Required:

1. Visual Studio Code (VS Code)
2. Google Chrome
3. HTML5
4. CSS3
5. JavaScript (ES6)

Theory:

JavaScript provides different types of functions that help organize code into reusable modules.

(a) Functions

Functions are reusable blocks of code that perform a specific task.

(b) Scope

Scope determines where variables can be accessed. JavaScript supports:

1. Global Scope
2. Function Scope
3. Block Scope

(c) Closures

A closure allows an inner function to access variables from its outer function even after the outer function has finished execution.

(d) Exception Handling

Exception handling is performed using the **try-catch** statement. The **try** block contains code that may generate an error, while the **catch** block handles the error without stopping the program.

Vehicle Number Validation

Accepts User Input Such As:

1. Vehicle Number

It Performs:

1. Validates the vehicle number format.
2. Displays a success message for a valid vehicle number.
3. Displays an error message for an invalid vehicle number using **try-catch**.

Result / Conclusion

The practical was completed successfully. The concepts of **functions, variable scope, closures, and exception handling using try-catch** were implemented. A vehicle number validation program was developed that accepts user input, validates the vehicle registration number, and displays appropriate success or error messages accurately using JavaScript.

Screenshot Attached